English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Groups 4/5, 18.12.2021 Test 1

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (2 points) According to the data used in this problem, about 2% of the population has COVID-19. Test manufacturers report that rapid tests are positive with probability 65% for infected people and 1% for healthy people.

1. (0.5 points) A randomly selected person tests positive. What is the probability they are infected?

   • Let $Inf = \{\text{the person is infected}\}$ and $Pos = \{\text{the test is positive}\}$. The required probability is

$$\mathbb{P}(Inf|Pos) = \frac{\mathbb{P}(Pos|Inf)\mathbb{P}(Inf)}{\mathbb{P}(Pos|Inf)\mathbb{P}(Inf) + \mathbb{P}(Pos|\overline{Inf})\mathbb{P}(\overline{Inf})} = \frac{65\% \times 2\%}{65\% \times 2\% + 1\% \times 98\%} \approx 57.02\%\,.$$

2. (0.5 points) In the setting of this problem, a negative test is needed for a green certificate. Suppose a person knows they are infected. How many tests on average are needed until a negative result[1]? What is the probability that two tests suffice to obtain the certificate?

   • The number of tests until the first positive result for an infected person is $X \sim Ge(p)$ with $p = 35/100$. Thus, on average, $100/35 \approx 2.86$ tests are needed until the first negative result, and the answer to the second part is $\mathbb{P}(X \le 2) = \mathbb{P}(X = 1) + \mathbb{P}(X = 2) = p + (1-p)p = 0.5775$.

3. (1 point) Consider two randomly selected people. What is the most likely pair (number infected, number of positive tests)? Calculate the correlation between these two counts, using approximations to accuracy 0.01.

   • Let X be the number infected, Y the number of positive tests, $p = 99/100$, $q = 65/100$ and $r = 2/100$. We calculate

$$\begin{aligned}
&\mathbb{P}(X=0,Y=0) = \mathbb{P}(X=0)\mathbb{P}(Y=0|X=0) = (1-r)^2p^2 \approx 0.94\\
&\mathbb{P}(X=0,Y=1) = (1-r)^2 \cdot 2p(1-p) \approx 0.02\\
&\mathbb{P}(X=0,Y=2) \approx 0.00\\
&\mathbb{P}(X=1,Y=0) = 2r(1-r)p(1-q) \approx 0.01\\
&\mathbb{P}(X=1,Y=1) = 2r(1-r)(pq + (1-p)(1-q)) \approx 0.03\\
&\mathbb{P}(X=1,Y=2) = 2r(1-r)q(1-p) \approx 0.00\\
&\mathbb{P}(X=2,Y=0) = \mathbb{P}(X=2)\mathbb{P}(Y=0|X=2) = r^2(1-q)^2 \approx 0.00\\
&\mathbb{P}(X=2,Y=1) = r^2 \cdot 2q(1-q) \approx 0.00\\
&\mathbb{P}(X=2,Y=2) = r^2q^2 \approx 0.00
\end{aligned}$$

   Hence the most likely outcome is two healthy people and zero positive tests, and the approximate joint distribution is

$X\backslash Y$	0	1	2	$\mathbb{P}(X=\cdot)$
0	0.94	0.02	0.00	0.96
1	0.01	0.03	0.00	0.04
2	0.00	0.00	0.00	0.00
$\mathbb{P}(Y=\cdot)$	0.95	0.05	0.00	

   ,

   Successively calculating gives

$$\begin{aligned}
&\mathbb{E}X \approx 0.04; && DX = \mathbb{E}X^2 - (\mathbb{E}X)^2 \approx 0.04 - (0.04)^2 \approx 0.04;\\
&\mathbb{E}Y \approx 0.05; && DY = \mathbb{E}Y^2 - (\mathbb{E}Y)^2 \approx 0.05 - 0.0025 \approx 0.05;\\
&\mathbb{E}XY \approx 0.03; && Cor(X,Y) = \frac{\mathbb{E}XY - \mathbb{E}X\mathbb{E}Y}{\sqrt{DX \cdot DY}} \approx \frac{0.03 - 0.04 \cdot 0.05}{\sqrt{0.04 \cdot 0.05}} \approx 0.63\,.
\end{aligned}$$

[1] For this problem, ignore the requirement to quarantine automatically after a positive result.

Problem 2. (1.5 points) A casino offers this game: roll a die 100 times and receive BGN 1 for each six. The entry fee is BGN 20.

- (0.5 points) Let X_i be the winnings from rolls $25i-24$ through $25i$ for $i = 1, 2, 3, 4$, and let X be the total winnings. What are the distributions of these random variables? Would you play?

  • $X_i \sim Bin(25, 1/6)$ and $X \sim Bin(100, 1/6)$. $\mathbb{E}X = 100/6 < 20$, so the game is not fair in terms of expected profit.

- (0.25 points) Let $Y = 4X_1$: instead of rolling 100 times, roll 25 times and multiply the winnings by 4. Compare the expectations and variances of X, Y, and briefly explain the result.

  • $\mathbb{E}X = \mathbb{E}X_1$, $DX = 500/36$, $D(4X_1) = 16DX_1 = 2000/36$. With more rolls, the deviation from the mean is naturally smaller.

- (0.75 points) The casino offers an alternative game: if the sum of 100 rolls exceeds 555, you win BGN 1000. The entry fee is BGN 10. An advertisement calls this a special Christmas offer and claims the winning probability exceeds 1/100, giving positive expected profit. Can you confirm or refute this?

  • Let Y_i be the value of roll i. We can write the total Y as $Y_1 + \cdots + Y_{100}$. We calculate $\mathbb{E}Y_1 = 7/2$ and $DY_1 = 35/12$. Since the Y_i are independent, $\mathbb{E}Y = 100 \cdot 7/2 = 350$ and $DY = 100 \cdot 35/12 = 875/3$. Thus, by Chebyshev's inequality,

$$\mathbb{P}(Y \geq 555) = \mathbb{P}(Y - \mathbb{E}y \geq 205) \leq \mathbb{P}(|Y - \mathbb{E}Y| \geq 205) \leq \frac{DY}{205^2} < 0.7\%\,,$$

  so the winning probability is strictly less than 1% and the advertisement is false.

Problem 3. (0.5 points) Two points are chosen at random on a unit segment. Call its endpoints left and right. What is the probability that the three resulting segment lengths increase from left to right?

• Let x, y be the distances of the two points from the left endpoint. If $y > x$, we need $x < y - x < 1 - y$, or $2x < y < (x + 1)/2$; similarly, if $x > y$, we need $2x - 1 < y < x/2$.

The final answer follows by elementary geometry or by

$$2\int_0^{1/3} \frac{x+1}{2} - 2x\,\mathrm{d}x = \int_0^{1/3} 1 - 3x\,\mathrm{d}x = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}\,.$$